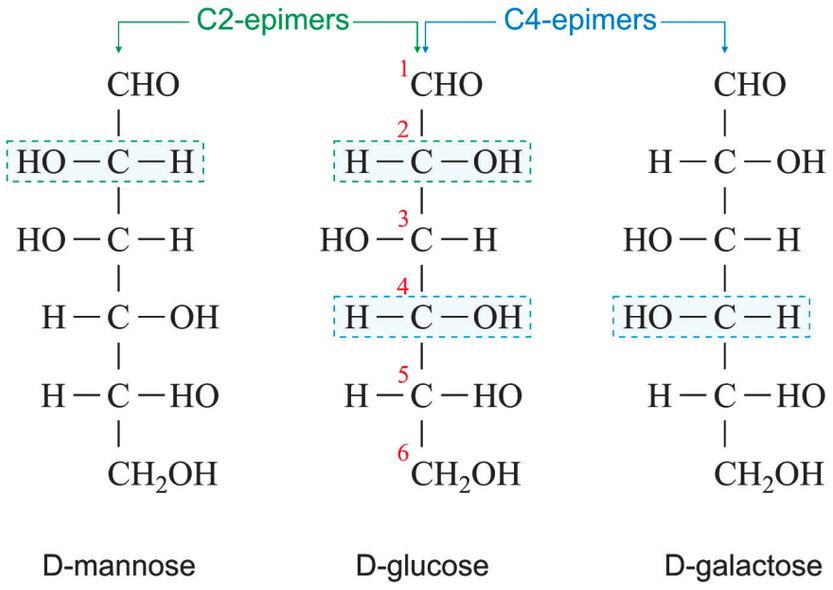
Epimers

D-sugars differing in configuration at a single asymmetric center are known as epimers. Thus D-glucose and D-galactose are epimers at C4 ; D- glucose and D-mannose are epimers at C2.



Anomers:

Sugars differing in configuration at the C1 asymmetric center are known as anomers. The C1–carbon is called as anomeric carbon. Thus α-D and β-D forms of glucose are anomers.

Mutarotation:

Mutarotation was discovered by chemisty dubrunfaut in 1814. Mutorotation is the change in optical rotation due to change in equilibrium between two anomers.

When D-glucose is crystallized at room temperature and a fresh solution is prepared, its specific rotation of polarized light is +112°; but after 12-18 hours it changes to +52.5°. If initial crystallization is taking place at 98° C and then solubilized, the specific rotation is found to be +19°, which also changes to +52.5° within few hours. This change in rotation with time is called mutarotation.

5.4 Properties of Glucose, Fructose and Galactose:

5.4.1 Glucose:

- Can be solid or liquid

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